Agentic AI frameworks

An agentic AI framework is a platform that makes it easy to develop and deploy agentic AI systems, which can seamlessly work with other agents, enterprise systems, and humans to achieve the required goals. It provides pre-built components and abstractions, tool integration, memory management, communication fabric, AI models for reasoning, and other basic infrastructure.

The components of an agentic AI framework can broadly be categorised as follows:

- ***Perception module:*** Includes devices that collect data from the environment; data processing software to clean up the raw data and extract useful information from it, like faces, names, and keywords; and natural language processing (NLP) to converse with stakeholders.
- ***Knowledge module:*** Stored information about the environment, past interactions, and experiences helps the system to understand the context and learn from it.
- ***Cognitive module:*** This module is the brain of the agentic AI system—an interplay of LLMs, ML, enterprise automation, and more. Here, the system understands its broader goal and current tasks, and deploys methods like chain-of-thought reasoning or heuristic decision trees to formulate a strategy and devise a series of steps to achieve it.
- ***Action module:*** The system now uses function calling to connect to AI agents, machines, data sources, web content, and APIs, to put its plan into action.
- ***Learning module:*** Feedback loops, RL, and other methods help the system to learn from its experiences and improve its functioning.
- ***Collaboration module:*** APIs, communication network, shared memory, and other elements that help the agentic AI integrate with existing enterprise systems, other AI systems, and human employees.
- ***Security module:*** Threat detection, data encryption, and other safeguards to protect the enterprise systems and data from external threats as well as from malfunctioning of AI systems.

“MANY PEOPLE DON’T UNDERSTAND THE IMPACT… SOME STILL THINK IT’S JUST ANOTHER TOOL. BUT AGENTIC AI WILL BRING A FUNDAMENTAL CHANGE IN HOW WE OPERATE. IT WILL CREATE NEW WAYS OF WORKING.”
— STUART BROWN, HEAD OF DIGITAL BUSINESS AT GUIDEHOUSE (IN A MEDIA INTERVIEW)

Agentic AI at work in different sectors

Enterprises around the world have already started deploying AI agents to improve productivity, reduce costs, enhance user experience, speed up decision-making, and ensure adaptability and scalability. Let us look at some of the major use cases for agentic AI.

- ***Customer service:*** Almost all sectors have deployed AI agents to implement intelligence and automation across the customer care fabric. Traditional AI is already used widely in customer service, but agentic AI takes it to the next level by not just answering customers’ queries but also helping resolve the issue. For example, an agent can simultaneously analyse customer sentiment, review order history, access company policies, and respond to customer needs based on these.
- ***Supply chain optimisation:*** Agentic AI can optimise supply chains by automating tasks, predicting demand, and managing inventory.
- ***Healthcare:*** Agentic AI helps to personalise patient care, automate tasks, and improve efficiency in healthcare. They can capture clinical notes during doctor-patient interactions, sort through vast mounds of data, and provide doctors with critical, actionable information, and handle other time-consuming tasks that burden doctors and healthcare professionals. On the other hand, AI agents can also provide round-the-clock support to patients, offering information on prescribed medication usage, appointment scheduling, and more.
- ***Content creation:*** Agentic AI can help create high-quality, personalised marketing content, freeing up the marketers’ time to focus on strategy and innovation.
- ***Data analysis:*** AI agents are good at analysing video, audio, images, text, or any other form of data to extract useful information—whether it is skimming through security footage to extract details of a crime, or monitoring high-altitude climbers using critical health parameters relayed from wearables and video feeds from drones, or monitoring financial transactions to detect and avert threats.
- ***Software development:*** Agentic AI can help manage the entire software development lifecycle, from system design to writing and debugging code and quality assurance. More commonly, software companies are using AI agents for repetitive coding tasks, enabling developers to focus on more complex challenges. In recent interviews, experts like Bill Gates, co-founder of Microsoft, Sam Altman, CEO of OpenAI, and Sridhar Vembu, founder and former CEO of Zoho Corporation, have